Employee Promotion Prediction

Ensemble Technique and Model Tuning

Contents

Executive Summary

- The HR department of the company can deploy the final model to identify whether an employee is eligible for promotion or not, more easily and efficiently it will help speed up the process. So, the HR department can utilize that time to identify and discuss appropriate appraisals for each employee.
- Our analysis showed that employees who are promoted have a higher average training score, the company should offer more training opportunities for the employees and inspire their employees to sign-up for more such training.
- Number of training for the majority of the employees is only 1, the company can offer training on new and interesting topics. The HR department can take feedback from the employees and decide which trainings can increase their productivity.
- We saw that only 2.3% of employees were given awards last year. The company can offer new and more opportunities for such awards to inspire employees to be better in their current roles.
- The HR department can work on collating more variables like average productive hours, average number of leaves in a month, etc. than can affect the productivity or promotion of an employee and can further improve the model performance.

Business Problem Overview and Solution Approach

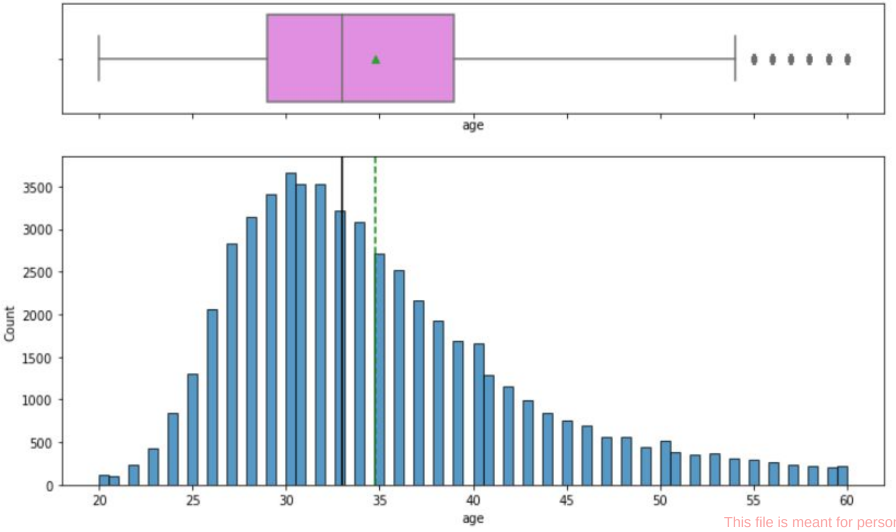
- Employee Promotion means the ascension of an employee to higher ranks, this aspect of the job is what drives employees the most. The ultimate reward for dedication and loyalty towards an organization and the HR team plays an important role in handling all these promotion tasks based on ratings and other attributes available

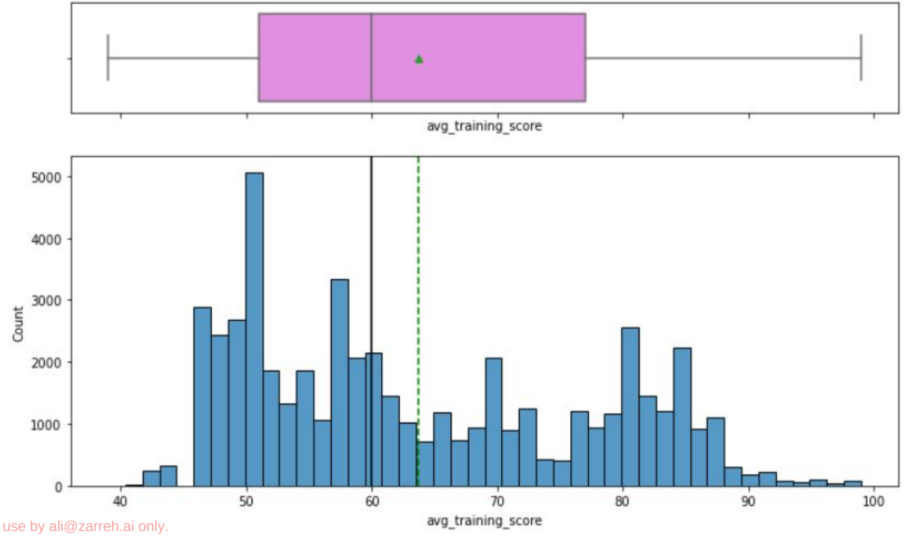
- The HR team in JMD company stored data on the promotion cycle last year, which consists of details of all the employees in the company working last year and also if they got promoted or not, but every time this process gets delayed due to so many details available for each employee - it gets difficult to compare and decide

- For the upcoming appraisal cycle, the HR team wants to utilize the stored data and leverage machine learning to make a model that will predict if a person is eligible for promotion or not. You, as a data scientist at JMD company, need to come up with the best possible model that will help the HR team to predict if a person is eligible for promotion or not

EDA Results

- The age distribution of employees is slightly skewed towards the right, with a majority of employees falling between the ages of 25 and 40. There are some outliers on the higher end of the age spectrum, as seen in the box plot.
- The distribution of average training scores is also slightly skewed towards the right, but no any outliers. The median score for average training is approximately 63, and most employees have a score below the median value.

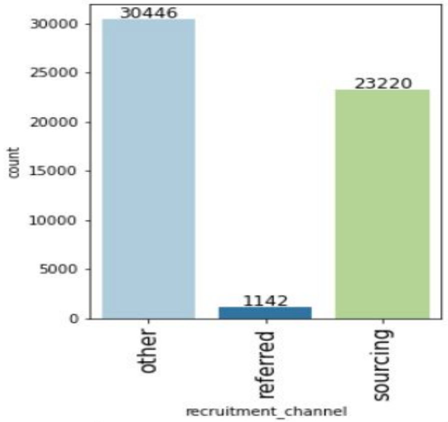


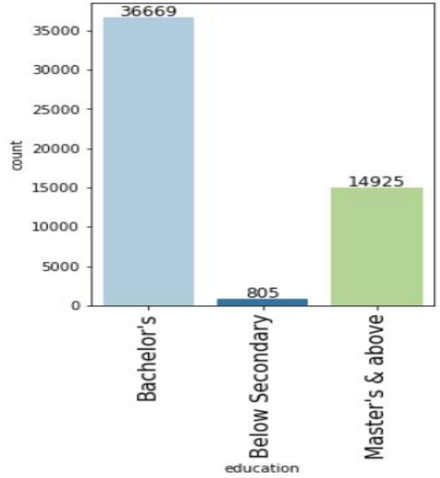


EDA Results

- Approx 45% of employees are hired through sourcing or referrals
- Approx 55% of employees are hired through other sources
- Very few employees i.e. approx 2% are hired through referrals. The company can give higher incentives to their current employees for referrals.

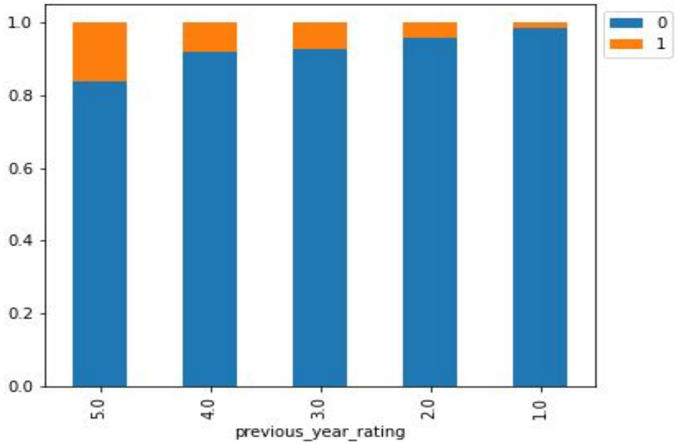
- Approx 67% of employees have a bachelor's level of education
- Approx 27% of employees have master's or above level of education
- There are very few employees i.e. 1.5% with the level of education below secondary



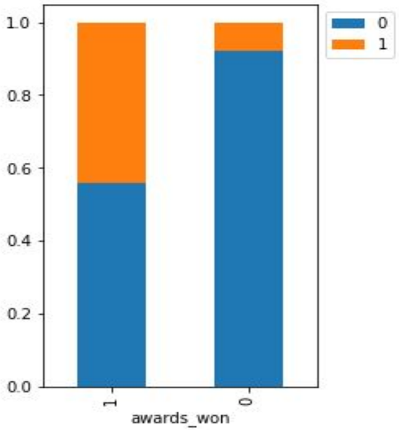


EDA Results

- The plot clearly shows that the higher the rating for the previous year higher is the chance of promotion in the current year.



- Employees who won an award are more likely to get promoted which can be expected as only the employee with excellent performance would get an award.

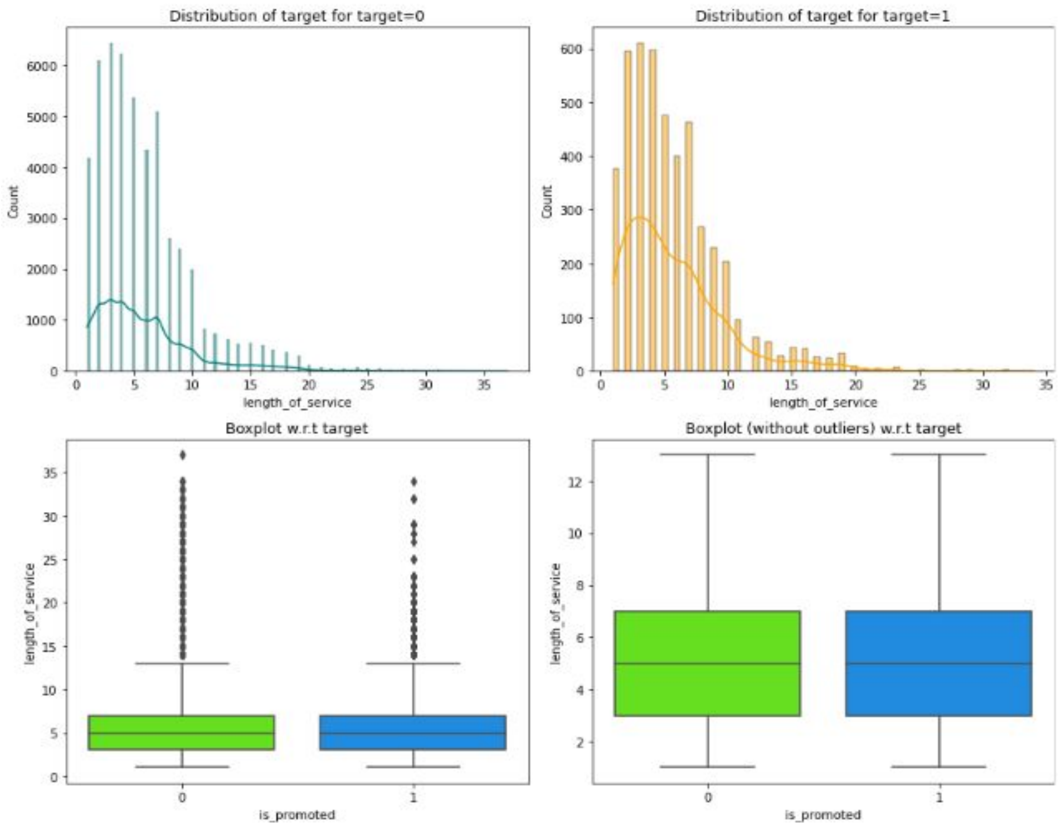


EDA Results

- Age and length of service have a high positive correlation which makes sense.
- The target variable is positively correlated with the avg_training_score and awards_won.
- There are no other variables with significant correlation

EDA Results

- There is not much difference in the distribution for promoted and non-promoted employees.



Data Preprocessing

- The data shared is a ciphered version containing 54808 observations.
- The data has various attributes or features that are associated with an employee, including their employee ID, department, region, level of education, gender, recruitment channel, number of training sessions attended, age, previous year's performance rating, length of service, awards won, average training score, and whether they have been promoted or not.
- There were few missing values in 3 columns The column 'education' has ~4.3% missing values, column 'previous_year_rating' has ~7.5% missing values and column 'avg_training_Score' has ~4.6% missing values. We imputed them using the most_frequent values for categorical columns and median for numerical columns and to avoid data leakage we imputed missing values after splitting train data into train and validation sets.
- ~91.5% of the employees are not promoted and only ~8.5% employees are promoted. The dataset is highly imbalanced so we tried undersampling and oversampling techniques to balance the data.

Comparing Model Performance

Training Performance

	XGBoost trained with Undersampled data	XGBoost trained with Original data	Gradient boosting trained with Undersampled data	Gradient boosting trained with Original data	AdaBoost trained with Undersampled data	AdaBoost trained with Original data
Accuracy	0.754	0.945	0.769	0.783	0.774	0.941
Recall	0.671	0.398	0.718	0.718	0.691	0.329
Precision	0.806	0.896	0.800	0.240	0.829	0.940
F1	0.753	0.761	0.768	0.614	0.773	0.728

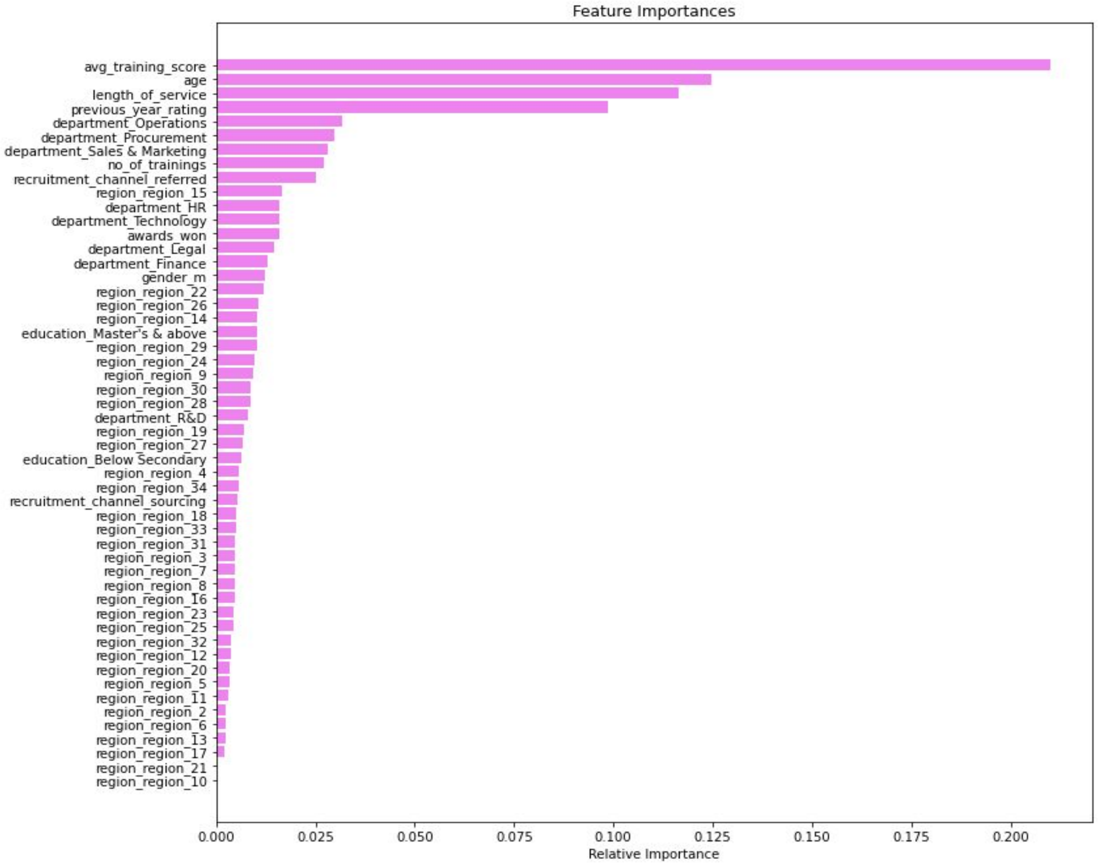
	XGBoost trained with Undersampled data	XGBoost trained with Original data	Gradient boosting trained with Undersampled data	Gradient boosting trained with Original data	AdaBoost trained with Undersampled data	AdaBoost trained with Original data
Accuracy	0.804	0.939	0.775	0.775	0.793	0.940
Recall	0.643	0.365	0.673	0.673	0.634	0.322
Precision	0.249	0.824	0.225	0.225	0.235	0.918
F1	0.622	0.737	0.601	0.601	0.610	0.723

Validation Performance

We can see that Adaboost trained with original data has best performance, followed by xgboost trained with original data

Feature Importance

- The average training score of the employee is the most important variable, for predicting an employee's promotion, followed by age, length of service and previous year rating.



Final Model Performance on test dataset

- We AdaBoosting model with original data.
- AdaBoost model performed well on unseen dataset.

Data	Accuracy	Recall	Precision	F1
Test	0.939	0.324	0.899	0.722

APPENDIX

Model Performance Summary (original data)

```
Cross-Validation Performance:

Bagging: 70.46454773316093
Random forest: 66.60680923823975
GBM: 69.53949501192146
Adaboost: 61.80957339408384
Xgboost: 72.01192446148467
dtree: 65.05089540323208

Validation Performance:

Bagging: 0.7060522382098885
Random forest: 0.6749629920498028
GBM: 0.6985500122041621
Adaboost: 0.6115403275532472
Xgboost: 0.7260409339776603
dtree: 0.6532891798794527
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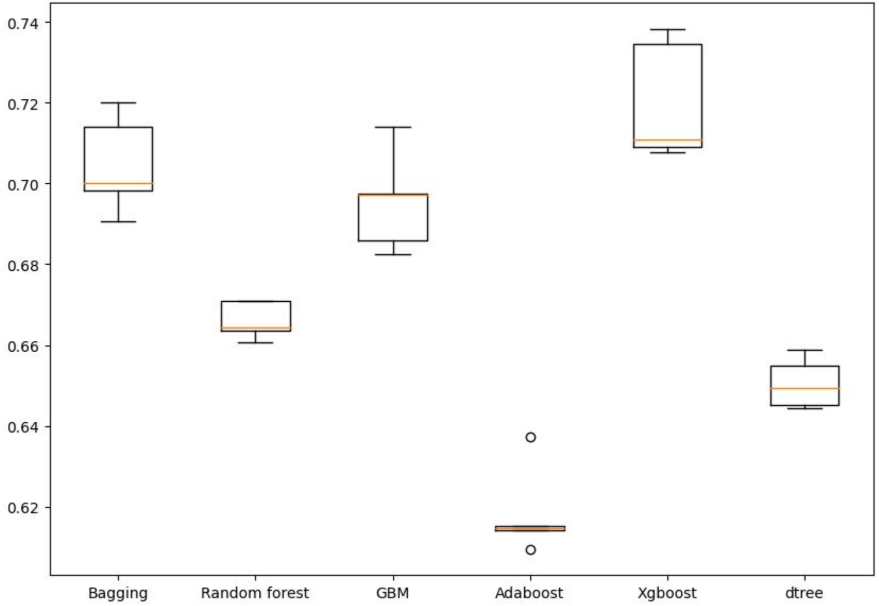

- Xgboost has the best performance followed by bagging classifier and gradient boosting classifier

Model Performance Summary (oversampled data)

```
Cross-Validation Performance:

Bagging: 94.32537064207025
Random forest: 95.3601586773084
GBM: 88.1161477265713
Adaboost: 87.69535552297489
Xgboost: 94.29621476166601
dtree: 91.96644300877225

Validation Performance:

Bagging: 0.6416862356660191
Random forest: 0.658534971390522
GBM: 0.624247849319147
Adaboost: 0.5727912561811965
Xgboost: 0.6835492221083692
dtree: 0.617107939820054
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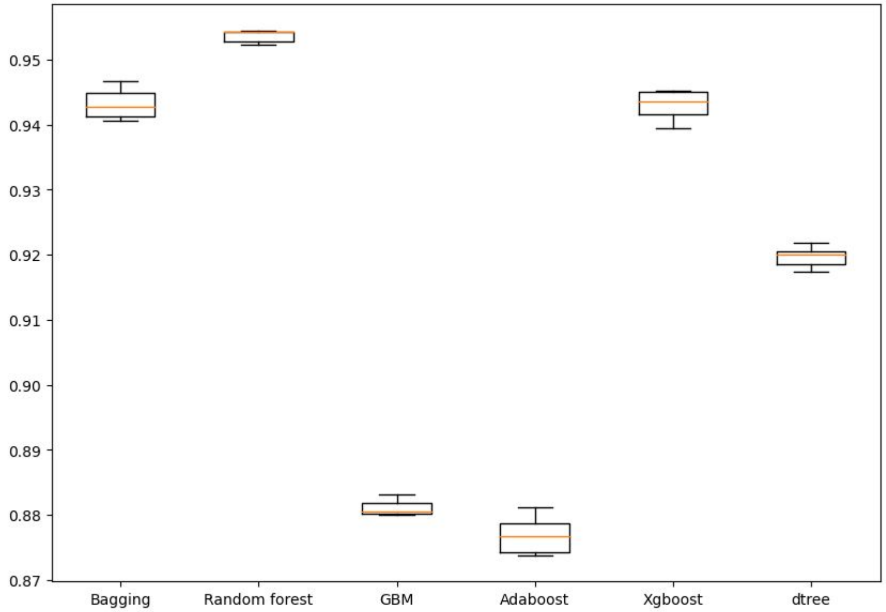

- The models built with oversampled data are highly overfitting
- Random forest is performing best here

Model Performance Summary (undersampled data)

```
Cross-Validation Performance:

Bagging: 68.6828971749485
Random forest: 70.71832081615916
GBM: 71.40936054357813
Adaboost: 69.67737064243286
Xgboost: 70.27797602444352
dtree: 64.9017237676074

Validation Performance:

Bagging: 0.5675127851045484
Random forest: 0.5749112015940396
GBM: 0.6232198174743587
Adaboost: 0.5724684187467689
Xgboost: 0.5812220134898979
dtree: 0.4979181761718061
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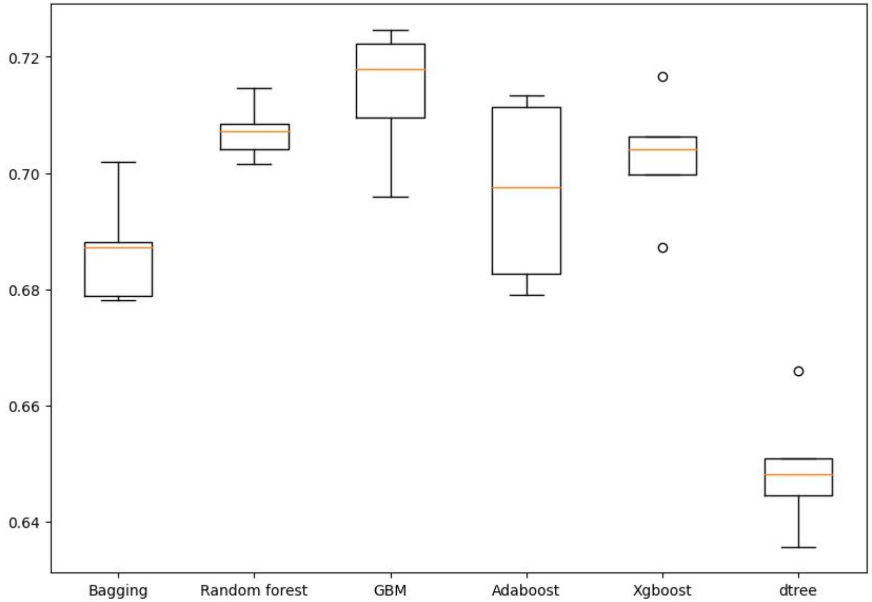

- Gradient boosting has the best performance followed by Adaboost and xgboost.
- Performance of Adaboost is consistent while xgboost has 2 outlier

Great Learning

Happy Learning !